through water toward an oncoming water wave, the wave approaches you more quickly; if you swim away from the wave, it approaches you more slowly. Similarly, if you move through the supposed aether toward or away from an oncoming light wave, the light wave's approach should, by the same reasoning, be faster or slower than 670 million miles per hour. In 1887, however, when Albert Michelson and Edward Morley measured the speed of light, time and time again they found exactly the same speed of 670 million miles per hour *regardless of their motion or that of the light's source.* All sorts of clever arguments were devised to explain these results. Maybe, some suggested, the experimenters were unwittingly dragging the aether along with them as they moved. Maybe, a few ventured, the equipment was being warped as it moved through the aether, corrupting the measurements. But it was not until Einstein had his revolutionary insight that the explanation finally became clear.

Relative Space, Relative Time

In June 1905, Einstein wrote a paper with the unassuming title "On the Electrodynamics of Moving Bodies," which once and for all spelled the end of the luminiferous aether. In one stroke, it also changed forever our understanding of space and time. Einstein formulated the ideas in the paper over an intense five-week period in April and May 1905, but the issues it finally laid to rest had been gnawing at him for over a decade. As a teenager, Einstein struggled with the question of what a light wave would look like if you were to chase after it at exactly light speed. Since you and the light wave would be zipping through the aether at exactly the same speed, you would be keeping perfect pace with the light. And so, Einstein concluded, from your perspective the light should appear as though it wasn't moving. You should be able to reach out and grab a handful of motionless light just as you can scoop up a handful of newly fallen snow.

But here's the problem. It turns out that Maxwell's equations do not allow light to appear stationary—to look as if it's standing still. And certainly, there is no reliable report of anyone's ever actually catching hold of a stationary clump of light. So, the teenage Einstein asked, what are we to make of this apparent paradox?